Functional Itô Calculus and Greeks of Path-Dependent Options

probabilistic representations and comparison with finite differences

Jeann Rocha

School of Applied Mathematics – FGV EMAp • **Advisor: Prof. Yuri Saporito**

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Motivation: beyond the Markov property

In mathematical finance, many contracts depend not only on the current asset price, but on the entire trajectory:

- **Asian options** — payoff on time averages;
- **Lookback options** — dependent on maxima/minima;
- **Realized volatility** — accumulated variance;
- **Functional volatility** — explicit path dependence.

Denoting the random variable of process X at time t by x_t and the respective path up to time t by X_t, i.e., $X_t = (x_s)_{0\le s\le t}$, consider the Black–Scholes model:

$$\mathrm{d}s_t = r s_t\,\mathrm{d}t + \sigma s_t\,\mathrm{d}w_t$$

In it, the price of a European option is a function only of time and the current price $V(t, s_t)$. A path-dependent derivative, on the other hand, pays based on the entire observed trajectory, so its price is a functional defined on the path space $V(t, S_t)$. The relevant dependence is no longer a point, but the whole history. It is this change of domain that the functional approach formalizes.

Path space and Dupire derivatives

Let $T > 0$ and

$$\Lambda = \bigcup_{t\in[0,T]} \Lambda_t$$

where Λ_t is the set of càdlàg paths on $[0, t]$, with $S_t \in \Lambda_t$.

Horizontal extension (*flat extension*). For $h > 0$,

$$S_{t,h}(u) = \begin{cases} S_t(u), & u \le t \\ S_t(t), & t < u \le t + h \end{cases}$$

The functional time derivative is:

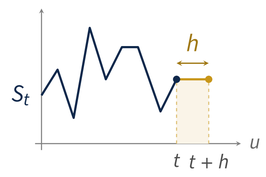
$$\Delta_t F(S_t) = \lim_{h\downarrow 0} \frac{F(S_{t,h}) - F(S_t)}{h}$$

Vertical perturbation (*bump*). For $\varepsilon \in \mathbb{R}$,

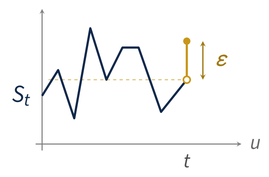
$$S_t^\varepsilon(u) = \begin{cases} S_t(u), & u < t \\ S_t(t) + \varepsilon, & u = t \end{cases}$$

The functional spatial derivative is:

$$\Delta_x F(S_t) = \lim_{\varepsilon\to 0} \frac{F(S_t^\varepsilon) - F(S_t)}{\varepsilon}$$



(a) Horizontal extension



(b) Vertical perturbation

Functional Itô formula

Let X be a continuous semimartingale and $F \in \mathbb{C}^{1,2}$ in the functional sense. The functional Itô formula states that

$$\mathrm{d}F(X_t) = \Delta_t F(X_t)\,\mathrm{d}t + \Delta_x F(X_t)\,\mathrm{d}x_t + \frac{1}{2}\Delta_{xx} F(X_t)\,\mathrm{d}\langle x\rangle_t$$

Compare with the classical formula:

$$\mathrm{d}f(t, x_t) = \partial_t f(t, x_t)\,\mathrm{d}t + \partial_x f(t, x_t)\,\mathrm{d}x_t + \frac{1}{2}\partial_{xx} f(t, x_t)\,\mathrm{d}\langle x\rangle_t$$

The difference is the *domain*: the classical formula acts on functions of finite-dimensional variables; the functional one on functionals defined in the path space.

Path dependence and Lie bracket

Jazaerli and Saporito introduce the bracket

$$\mathcal{L}F = [\Delta_t, \Delta_x](F) = \Delta_t\Delta_x F - \Delta_x\Delta_t F$$

which measures the non-commutativity between time and spatial perturbations. Geometrically, it compares the operations of vertically perturbing a path and horizontally extending it.



Commutativity diagram

Path dependence classification

- $\mathcal{L}F = 0$: locally weakly path-dependent functional;
- $\mathcal{L}F \neq 0$: strongly path-dependent functional.

This distinction allows obtaining probabilistic representations of the Greeks, with Monte Carlo weights adapted to the dependence structure of the contract.

Financial application: pricing and Greeks

Consider an asset with dynamics, under the risk-neutral measure $\mathbb{Q}$, given by

$$\mathrm{d}s_t = r\,s_t\,\mathrm{d}t + \sigma(S_t)\,\mathrm{d}w_t^{\mathbb{Q}}$$

For a payoff $g : \Lambda_T \to \mathbb{R}$, the no-arbitrage price is

$$V(S_t) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}[g(S_T) \mid S_t] = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}\left[g(S_T^{S_t})\right]$$

where the path $S_T^{S_t}$ coincides with S_t up to t and evolves according to the model. By the functional Feynman–Kac theorem, V satisfies the functional PDE

$$\Delta_t V(S_t) + r\,s_t\,\Delta_x V(S_t) + \tfrac{1}{2}\sigma^2(S_t)\,\Delta_{xx} V(S_t) - rV(S_t) = 0$$

The Greeks are functional derivatives of the price:

$$\Delta = \Delta_x V(S_t), \qquad \Gamma = \Delta_{xx} V(S_t), \qquad \text{Vega} = \Delta_\sigma V(S_t)$$

The goal is to obtain probabilistic representations of these quantities, allowing Monte Carlo estimation.

Delta representation by weighted expectation

If V is locally weakly path-dependent, we have:

$$\Delta_x V(S_0) = \mathbb{E}\left[g(S_T)\,\frac{1}{T}\int_0^T \frac{z_t}{\sigma(S_t)}\,\mathrm{d}w_t \,\middle|\, S_0\right]$$

where z_t is the tangent process associated with the underlying asset dynamics given by $\mathrm{d}z_t = \Delta_x\sigma(S_t) z_t\,\mathrm{d}w_t$, $z_0 = 1$.
When the functional exhibits strong dependence, Delta acquires a correction term:

$$\Delta_x V(S_0) = \mathbb{E}\left[g(S_T)\,\frac{1}{T}\int_0^T \frac{z_t}{\sigma(S_t)}\,\mathrm{d}w_t \,\middle|\, S_0\right] + \mathbb{E}\left[\frac{1}{T}\int_0^T\int_0^t \mathcal{L}V(S_s)\,z_s\,\mathrm{d}s\,\mathrm{d}t \,\middle|\, S_0\right]$$

The second term is a path correction that captures the impact of strong dependence on sensitivity.
These representations allow estimating Delta via Monte Carlo without numerically differentiating the price.

Finite difference method

Delta via central finite differences perturbs the initial price s_0 and reuses the same paths via common random numbers. Denoting by S^{s_0} the path generated from s_0:

$$\Delta \approx \frac{V(S^{s_0+h}) - V(S^{s_0-h})}{2h}.$$

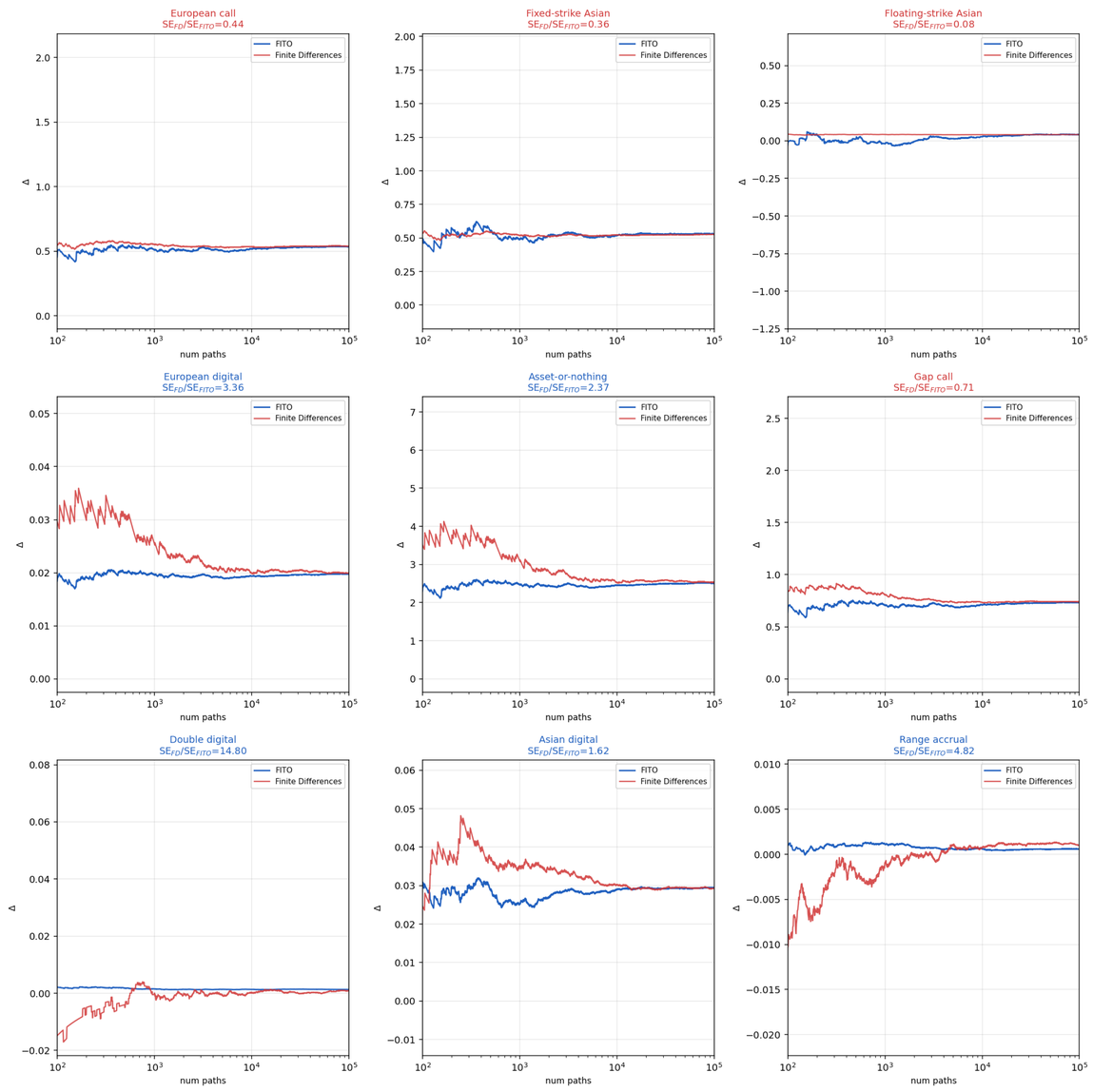
The parameter h controls the bias-variance trade-off.

Comparison of methods

The table compares the weighted expectation via FITO and finite differences on classical payoffs. The last column, $\mathrm{SE}_{\mathrm{FD}}/\mathrm{SE}_{\mathrm{FITO}}$, summarizes the relative efficiency: values > 1 indicate that FITO achieves a lower standard error with the same number of paths.

Option	Payoff	Weighted expectation			Finite differences			Efficiency
		Δ	SE	t (s)	Δ	SE	t (s)	$\mathrm{SE}_{\mathrm{FD}}/\mathrm{SE}_{\mathrm{FITO}}$
European call	$(s_T - K)^+$	0,5342	0,0042	0,0014	0,5391	0,0019	0,0030	0,44
Fixed-strike Asian	$(\bar{s} - K)^+$	0,5298	0,0049	0,0013	0,5250	0,0017	0,0562	0,36
Floating-strike Asian	$(s_T - \bar{s})^+$	0,0408	0,0028	0,0013	0,0410	0,0002	0,0578	0,08
European digital	$\mathbf{1}_{\{s_T>K\}}$	0,0197	0,0001	0,0015	0,0199	0,0003	0,0033	**3,36**
Asset-or-nothing	$s_T\,\mathbf{1}_{\{s_T>K\}}$	2,5082	0,0131	0,0013	2,5331	0,0310	0,0050	**2,37**
Gap call	$(s_T - K_1)\,\mathbf{1}_{\{s_T>K_2\}}$	0,7316	0,0050	0,0013	0,7385	0,0036	0,0060	0,71
Double digital	$\mathbf{1}_{\{K_l<s_T<K_u\}}$	0,0013	0,0000	0,0012	0,0008	0,0004	0,0056	**14,80**
Asian digital	$\mathbf{1}_{\{\bar{s}>K\}}$	0,0294	0,0002	0,0014	0,0293	0,0004	0,0570	**1,62**
Range accrual	$\mathbf{1}_{\{L\le s_{t_j}\le U\}}$	0,0006	0,0000	0,0014	0,0010	0,0002	0,0703	**4,82**

Model: Black–Scholes ($r = 0$, $\mathrm{d}s_t = \sigma s_t\,\mathrm{d}W_t$), $s_0 = 100$, $\sigma = 20\%$, $T = 1$, $n = 10^5$ paths, 252 steps, perturbation $h = 1\% \cdot s_0$.
Weights: terminal $w_T = W_T/(T\sigma s_0)$; local $w_t = W_{t_1}/(t_1\sigma s_0)$ with $t_1 = 0{,}2$; $\bar{s} = (T - t_1)^{-1}\int_{t_1}^T s_u\,\mathrm{d}u$. t: Greek computation time.



Convergence of Δ estimates as a function of the number of paths. Blue: FITO | Red: Finite Differences.

Analysis. Finite differences (FD) prove more efficient for smooth payoffs (European and Asian), where common random numbers cancel the noise. However, FITO is drastically superior for discontinuous payoffs (digital and barrier), reducing the standard error by up to 15 times, as the Malliavin weight smooths the variance that explodes in FD near discontinuities. Furthermore, FITO obtains the Greek in the same simulation as the price, while FD requires extra simulations.

Perspectives

- **Geometry.** Càdlàg paths as points of an infinite-dimensional manifold; Δ_t, Δ_x as vector fields and $\mathcal{L}F$ as their Lie bracket.
- **Functional analysis.** Adjoints Δ_t^*, Δ_x^*, Δ_{xx}^*, integration by parts, weak and Galerkin formulations in the path space.
- **Beyond finance.** Non-Markovian Langevin dynamics (physics), sequential and functional data analysis (statistics), history-dependent epidemic models (epidemiology).

References

1. JAZAERLI, S.; SAPORITO, Y. F. *Functional Itô calculus, path-dependence and the computation of Greeks*. Stochastic Processes and their Applications, v. 127, n. 12, p. 3997–4028, 2017.
2. DUPIRE, B. *Functional Itô Calculus*. Bloomberg Quantitative Research, 2009.
3. CONT, R.; FOURNIÉ, D.-A. *Functional Itô calculus and stochastic integral representation of martingales*. The Annals of Probability, 2013.
4. CONT, R.; FOURNIÉ, D.-A. *A functional extension of the Itô formula*. Comptes Rendus Mathématique, 2010.

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